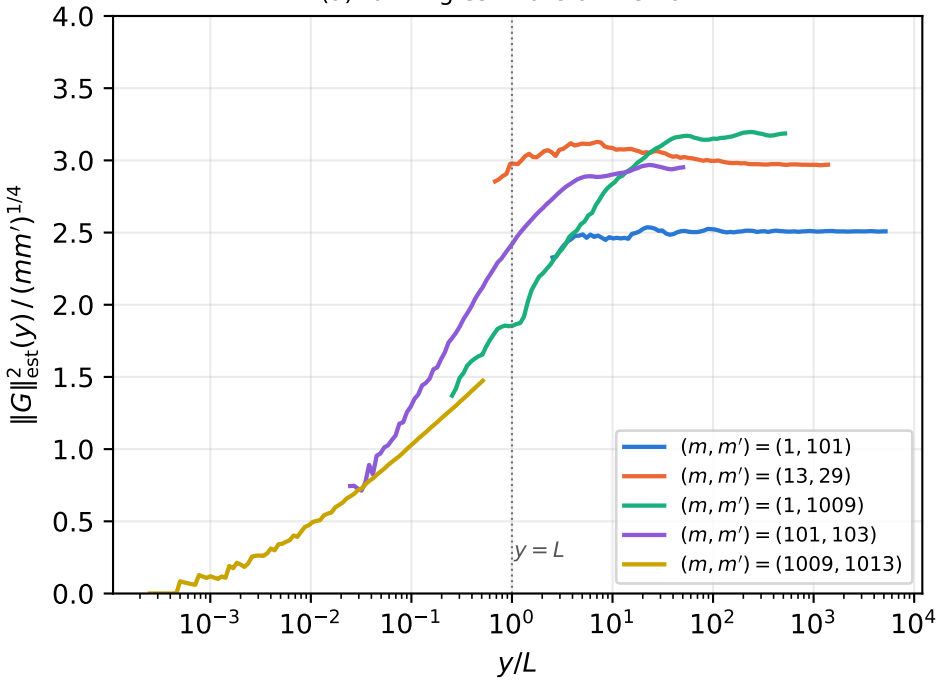
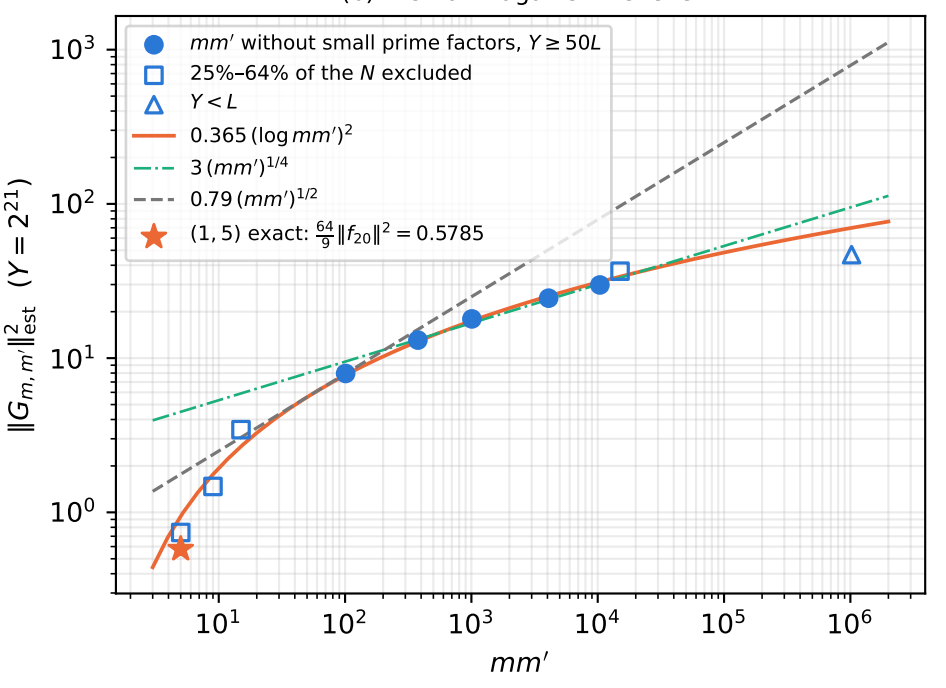

(a) running estimate of the norm
$\|G\|^2_{\mathrm{est}}(y)\,/\,(mm')^{1/4}$
y/L
$y = L$
$(m, m') = (1, 101)$
$(m, m') = (13, 29)$
$(m, m') = (1, 1009)$
$(m, m') = (101, 103)$
$(m, m') = (1009, 1013)$


(b) the norm against the level
$\|G_{m,m'}\|^2_{\mathrm{est}}$ $(Y = 2^{21})$
mm'
mm' without small prime factors, $Y \geq 50L$
25%–64% of the N excluded
$Y < L$
$0.365\,(\log mm')^2$
$3\,(mm')^{1/4}$
$0.79\,(mm')^{1/2}$
(1, 5) exact: $\frac{64}{9}\|f_{20}\|^2 = 0.5785$